INCH-POUND

MIL-R-83726/23D
19 September 2005
SUPERSEDING
MIL-R-83726/23C(USAF)
22 December 1994

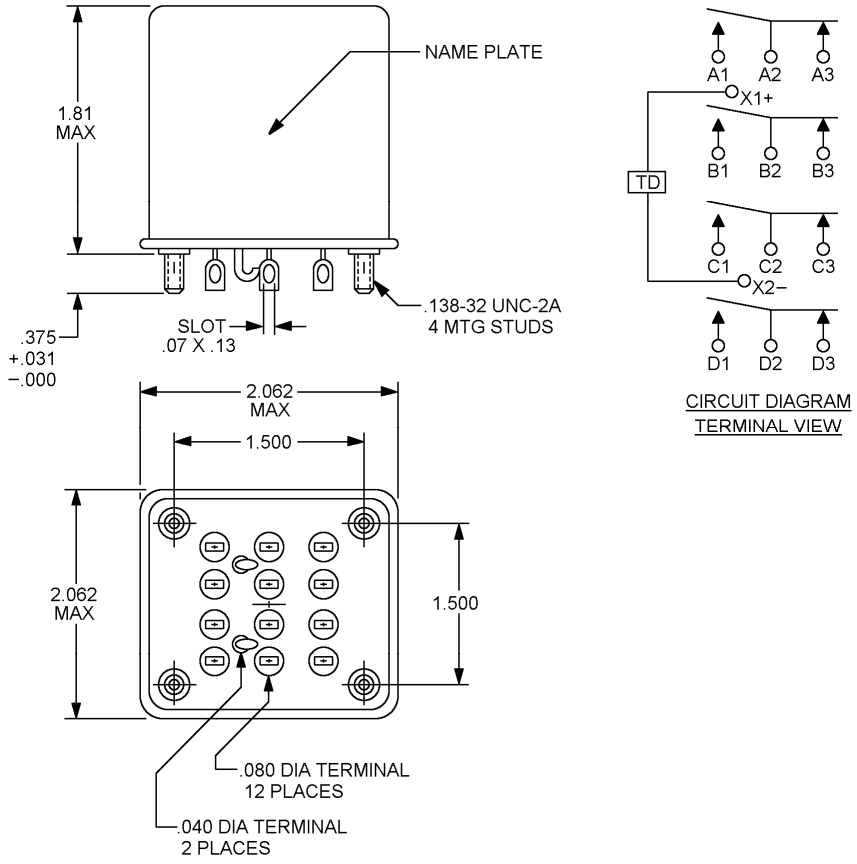
DETAIL SPECIFICATION SHEET

RELAYS, HYBRID, TIME DELAY (ON RELEASE), TYPE IIB, CLASS B, 10 AMPERES, 4PDT, HERMETICALLY SEALED

Inactive for new design after 10 November 1993.
No superseding document.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

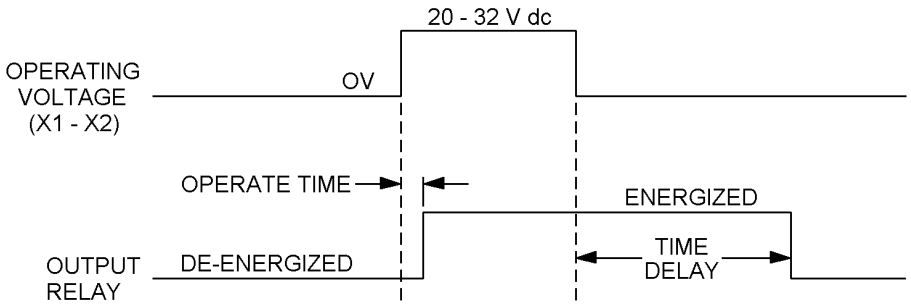
The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-83726.



Inches	mm
.031	0.79
.040	1.02
.070	1.78
.080	2.03
.130	3.30
.138	3.51
.375	9.53
1.500	38.10
1.810	45.97
2.062	52.37

FIGURE 1. Dimensions and configurations.

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TIMING DIAGRAM

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.03 (0.8 mm) for two place decimals.
4. Terminal numbers shall not appear on the relay header. There shall be affixed to the relay a permanent, legible circuit diagram that identifies each terminal location specified.

FIGURE 1. Dimensions and configurations - Continued.

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TABLE I. Dash numbers and time delay characteristics. 1/

Dash number	Release time delay (seconds) ±10%	Maximum operating time (ms)	Dash number 2/	Release time delay (seconds) ±10%	Maximum operating time (ms)
1000	2/ .100	50	1602	16	100
5000	2/ .500	50	1802	18	100
7500	2/ .750	50	2002	20	100
1001	1	50	2202	22	100
2001	2	50	2502	25	100
3001	3	50	3002	30	100
4001	4	50	3502	35	100
5001	5	50	4002	40	100
6001	6	50	4502	45	100
7001	7	50	5002	50	100
8001	8	50	5502	55	100
9001	9	50	6002	60	100
1002	10	50	6502	65	100
1202	12	100	7002	70	100
1402	14	100	7502	75	100

1/ Additional time delay relays within the 0.1 to 75 seconds delay range are available. To establish Part or Identifying Number (PIN's) not listed in table I, see "PIN" herein. Time delay accuracy applies to all combinations of operating temperature and voltage.

2/ Add ±10 milliseconds to ±10 percent tolerances.

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REQUIREMENTS:

Contact data:

Configuration: 4PDT.

Life/load ratings (relay case grounded) (See table II).

TABLE II. Life/load ratings.

Type of load	Life (cycles)	Amperes	
		28 V dc	115/200 V rms 1 and 3 phase, 400 Hz
Resistive	100,000	10	10
Inductive	20,000	8	8
Motor	100,000	4	4
Lamp	100,000	2	2
Reduced current resistive	400,000	2.5	2.5

Contact voltage drop:

Initial: 0.150 volt.

After life tests: 0.175 volt.

Minimum current: Applicable in accordance with MIL-PRF-6106.

Contact bounce: 1 millisecond (maximum).

Overload:

DC: 40 amperes.

AC: 60 amperes.

Rupture:

DC: 50 amperes.

AC: 80 amperes.

Input data:

Duty rating: Continuous.

Maximum voltage: 32 V dc.

Rated voltage (over temperature range): 28 V dc.

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Minimum voltage: 20 V dc.

Minimum voltage (high-temperature test): 20 V dc.

Minimum voltage (continuous current test): 20 V dc.

Peak current (-55°C to +100°C): 2.5 amperes for 100 milliseconds (maximum).

Steady-state dc current: 0.060 ampere (maximum).

Release time delay: See table I.

Recycle time: 1/

Maximum operate time: See table I.

Electrical data:

Insulation resistance at 500 V dc: 2/

Initial: 1,000 megohms (minimum).

After life or environmental tests: 500 megohms (minimum).

Dielectric withstanding voltage (sea level):

	Initial	After life tests
Input (X1 - X2) 2/	1,000 V rms	1,000 V rms
All other points.	1,250 V rms	1,000 V rms

Dielectric withstanding voltage (altitude) (80,000 feet): 2/

Input (X1 - X2): 350 V rms. 2/

All other points: 350 V rms.

Transients: In accordance with MIL-STD-704 for 28 volts dc system (figure 11).

1/ Recycle time is the same as operate time in table I. It is defined as the minimum time that operate voltage must be applied to the input terminals to assure that a new release timing cycle can be completed within the specified timing tolerance.
2/ Input terminals X1 and X2 shall be connected together during this test.

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Environmental data:

Temperature range (operating): -55°C to +100°C.

Maximum altitude rating: 80,000 feet.

Shock g-level: 50 g's, 1/2 sine, 3 axes.

Duration: 11 ±1 milliseconds.

Maximum duration contact opening: 10 microseconds.

Vibration (sinusoidal):

G-level: 20 g's

Frequency range: 10 Hz to 3,000 Hz.

Vibration (random): Applicable in accordance with MIL-STD-202, method 214, test condition IG.

Power spectral density: 0.4 g^2/Hz.

Frequency range: 50 Hz to 2,000 Hz.

Duration: 15 minutes each plane.

Acceleration: 20 g's in any axis.

Seal: MIL-STD-202, method 112, condition B.

Humidity: 95 percent relative humidity.

Physical data:

Dimensions and configuration: See figure 1.

Terminations: Solder hook.

Terminal strength: 3 ±0.5 pounds pull.

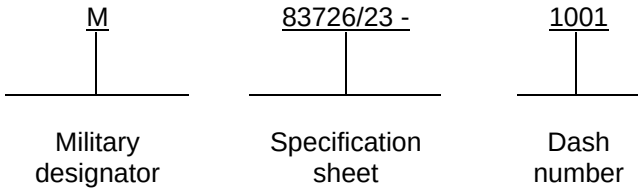
Weight: 0.75 pound (maximum).

Marking: See MIL-PRF-83726. In addition, relays shall be marked with the ESDS identifier as specified in MIL-STD-1285.

Electrostatic discharge (ESD) control program: Applicable (see MIL-PRF-83726).

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Part or Identifying Number (PIN): Consists of the prefix M83726/23-, a four-digit dash number (expressed in milliseconds) as follows:



Examples

The first three digits of the dash number are significant; the fourth digit is the number of zeros to follow the first three digits. The time delay is expressed in milliseconds and converted to seconds.

M83726/23 - 1000 - 0.100 second time delay.
M83726/23 - 6001 - 6 seconds time delay.

Referenced documents. In addition to MIL-PRF-83726, this document references the following:

MIL-STD- 202 MIL-STD-704 MIL-STD-1285 MIL-PRF-6106

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5945-1316)

Review activities:
Air Force - 99

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